

Data Profiling and QA Tools



Data profiling helps us understand the quality of data before doing any analysis. It helps us find issues like missing values, errors, duplicates, and incorrect data types.

1. What kind of data do we have before doing any analytics?

Raw data

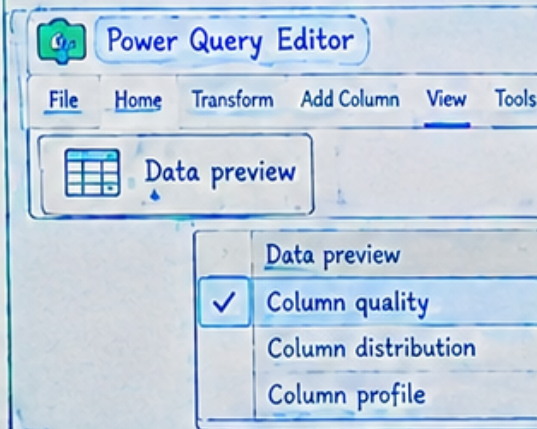
ID	Name	Age	Email	Sales
1	Kajal	25	kajal@abc.com	1000
2	Rohit	30	rohit@abc.com	2000
3	Sneha	-	sneha@abc.com	1500
4	Amit	28	NULL	1800
5	Pooja	35	pooja@abc.com	0

We may have:

- Correct / valid data
- Missing values (blank/empty)
- Incorrect data (wrong format)
- Duplicates
- Errors (e.g. text in number column)
- Inconsistent values (e.g. different spellings, case, etc.)

So, we need to assess and clean the data before analysis.

2. Do data assessment using tool – Power Query Editor



1. Open Power Query Editor
2. Go to View tab
3. Click Data preview
4. Select Column quality

This shows the quality of each column in your data.

3. Column Quality – What it shows?

1. Valid Data

- Correct and usable data.
- Follows the expected format and rules.

Example:

Name	Count
Kajal	45
Rohit	38
Sneha	52

2. Error Data

- Data that cannot be converted or doesn't match the data type.
- May show errors like #N/A, #DIV/0!, etc.

Example:

Sales	Count
1200	98
#N/A	2
0/0	1

3. Empty Data

- Blank or missing values.
- Can affect analysis and calculations.

Example:

Email	Count
kajal@abc.com	40
(blank)	12
rohit@abc.com	35



On hover, we get suggestions to remove empty or error values (e.g. "Remove Empty", "Remove Errors", "Replace Values").

4. Column Distribution – What it shows?

- Shows the spread of values in a column.
- Helps us understand the data pattern (e.g. frequency of each value).
- Useful for finding outliers, skewness, or dominant values.

Example:

Country	Count
USA	40
India	30
UK	20
Canada	10

Column Distribution (Country)



It shows how many times each value appears in the column.

5. Column Profile – What it shows?

Gives a detailed summary of a column, including:

- Data type
- Distinct count
- Minimum / Maximum
- Average (for numeric data)
- Empty count
- Error count

Example: (Sales column)

Column profile	
Data type	Whole number
Distinct count	45
Minimum	0
Maximum	5000
Average	1250
Empty count	3
Error count	1

Also shows:

Value distribution



Top values

2000	5
1500	4
1000	4

6. Summary – What we get from Data Profiling?

- ✓ Data quality insights
- ✓ Find and fix errors, missing values, duplicates
- ✓ Understand data distribution
- ✓ Better and accurate analysis
- ✓ Save time in data cleaning



Key Takeaway:

Data profiling in Power Query helps us understand our data, find issues early, and prepare clean, reliable data for meaningful analytics.



Text tool – Transformation

What is the use of TEXT tool in Power BI ?

- The TEXT tool is used to clean, format and manipulate text data in Power BI (Power Query).
- It helps in standardizing text, splitting text into parts, extracting useful information, merging columns, etc.
- Useful when your data has inconsistent formats, extra spaces, mixed case or needs to be combined with other columns.

Home	Transform	Add Column	View	Tools	Help
 Text	 Format	 Extract	 Parse	 Merge Columns	

Transform → Text Tools

Example – Raw Data

Name	Code	City
john doe	a123	new york
Priya Sharma	b456	mumbai
RAHUL KUMAR	c789	bangalore

→ Using text tool we can clean, format and split this data.

1 Format

Used to change the case or add/remove text from the existing column.

Options :

- Lowercase – all letters to small case
- Uppercase – all letters to capital case
- Capitalize each word – first letter of each word capital
- Trim – remove leading & trailing spaces
- Clean – remove extra spaces, line breaks etc.
- Add prefix – add text at the beginning
- Add suffix – add text at the end

Example (Format)

Original	After Format (e.g. Capitalize each word)
john doe	John Doe
priya	Priya
RAHUL KUMAR	Rahul Kumar

"If we add prefix "Mr." to Name column:

Original	With Prefix
John Doe	Mr. John doe
Priya Sharma	Mr. Priya Sharma

2 Split Column

Used to split text into multiple columns.

Options :

- By delimiter – e.g. split by comma (,), space, etc.
- By number of characters – e.g. first 3 characters
- By position – e.g. at position 3 (before/after)
- By lowercase to uppercase – change case while splitting

Example (Split by delimiter ,)

Original	After Split
John, Doe, Sales	John Doe Sales

3 Extract

Used to pull specific parts of text.

Options :

- Len – get length of text
- First – first n characters
- Last – last n characters
- Text before delimiter – e.g. before '@'
- Text after delimiter – e.g. after '@'
- Text between delimiters – e.g. between '(' and ')'
- Range – extract characters by position

Example (Extract)

Function	Result (for "abc@xyz.com")
Len	11
First 3	abc
Last 3	com
Text before @	abc
Text after @	xyz.com
Text between (.)	(none)
Range (1,3)	bc

4 Parse

Used to extract structured data from text (like JSON, XML, or custom text).

Example :

- From a text value like "City: New York; Zip: 10001" we can parse City and Zip into separate columns.

5 Merge Columns

Used to combine two or more columns into one column.

Example :

- First Name + Last Name → Full Name

First Name	Last Name	Full Name
John	Doe	John Doe

Important Note

Instead of using Transform directly, use Add Column → Merge Columns to create a new column.

- Transform will modify the existing column, which may not be useful if you want to keep the original data.

Numeric Tools – Power BI

What are Numeric Tools ?

- Numeric tools are used to perform calculations and analysis on numbers in Power BI (Power Query / DAX).
- They help us get useful insights like totals, averages, trends, rounding, and more.
- These tools are grouped into categories like Statistics, Standard, Scientific, Trigonometry, Rounding and Information.

2 Standard

Performs basic arithmetic and percentage calculations.

Function	Use	Example (A=100, B=20)
Add	Adds two numbers.	$100 + 20 = 120$
Multiply	Multiplies two numbers.	$100 * 20 = 2000$
Subtract	Subtracts second from first.	$100 - 20 = 80$
Divide	Divides two numbers.	$100 / 20 = 5$
Integer-Divide	Returns whole number (no decimal).	$100 // 20 = 5$
Modulo	Returns remainder.	$100 \% 30 = 10$
Percentage	Converts to percentage.	$20\% = 20/100 = 0.2$
Percent of	Finds a percentage of a value.	$20\% \text{ of } 200 = 200 * 0.2 = 40$

Example: If Total = 500, Discount = 10%

- Subtract = $500 - 50 = 450$
- Multiply = $500 * 0.10 = 50$
- Percent of (10%) = $500 * 0.10 = 50$

4 Trigonometry

Used for angle and triangle related calculations (in radians or degrees).

Function	Use	Example (angle = 30°)
SIN	Sine of an angle.	$\text{SIN}(30^\circ) = 0.5$
COS	Cosine of an angle.	$\text{COS}(30^\circ) = 0.866$
TAN	Tangent of an angle.	$\text{TAN}(30^\circ) = 0.577$
ASIN	Inverse sine.	$\text{ASIN}(0.5) = 30^\circ$
ACOS	Inverse cosine.	$\text{ACOS}(0.866) = 30^\circ$
ATAN	Inverse tangent.	$\text{ATAN}(0.577) = 30^\circ$

Example: If angle = 45°

- $\text{SIN}(45^\circ) = 0.707$
- $\text{COS}(45^\circ) = 0.707$
- $\text{TAN}(45^\circ) = 1$

Note: In Power BI, angles are usually in radians (unless specified).

1 Statistics

Used to find key statistical values from a set of numbers.

Function	What it does	Example (on data: 10,20,30,40,50)
Sum	Adds all values.	$\text{SUM} = 150$
Min	Returns the smallest value.	$\text{MIN} = 10$
Max	Returns the largest value.	$\text{MAX} = 50$
Median	Middle value when data is sorted.	$\text{MEDIAN} = 30$
Average	Mean of the values.	$\text{AVERAGE} = 30$
Std Deviation	Shows how spread out the data is.	$\text{STDEV} = 15.81$
Count Values	Counts non-blank values.	$\text{COUNT} = 5$
Count Distinct	Counts unique values.	$\text{COUNT DISTINCT} = 5$

Example: If Sales = {10, 20, 30, 40, 50}

- Sum = 150
- Min = 10
- Max = 50
- Avg = 30
- Median = 30
- Std Dev ≈ 15.81
- Count = 5
- Distinct Count = 5

3 Scientific

Used for advanced mathematical operations.

Function	Use	Example
Power	Raises a number to a power.	$\text{POWER}(2,3) = 8$
Square Root	Finds square root.	$\text{SQRT}(25) = 5$
Log	Natural/logarithm base 10.	$\text{LOG}(100) = 2$ (LOG10)
Exp	e raised to a power.	$\text{EXP}(1) = 2.718$
Factorial	Product of all positive integers up to n.	$\text{FACTORIAL}(5) = 120$

Example: If a = 2

- $\text{POWER}(2,3) = 8$
- $\text{SQRT}(16) = 4$
- $\text{LOG}(100) = 2$
- $\text{EXP}(1) = 2.718$
- $\text{FACTORIAL}(5) = 120$

5 Rounding

Used to round numbers to a specified place or direction.

Function	Use	Example (n=2.6)
Round Up	Always rounds up (away from zero).	$\text{ROUNDUP}(2.6,0) = 3$
Round Down	Always rounds down (towards zero).	$\text{ROUNDDOWN}(2.6,0) = 2$
Round	Rounds to nearest value.	$\text{ROUND}(2.6,0) = 3$

Example: If number = 2.345

- Round Up (2) = 2.35
- Round Down (2) = 2.34
- Round (2) = 2.35

6 Information

Used to check properties of a number.

Function	Use	Example
Is Even	Checks if number is even.	$\text{IS.EVEN}(4) = \text{TRUE}$ $\text{IS.EVEN}(5) = \text{FALSE}$
Is Odd	Checks if number is odd.	$\text{IS.ODD}(5) = \text{TRUE}$ $\text{IS.ODD}(4) = \text{FALSE}$
Sign	Returns the sign of a number (positive, negative, or zero).	$\text{SIGN}(10) = 1$ $\text{SIGN}(-5) = -1$ $\text{SIGN}(0) = 0$

Example: If number = -8

- Is Even = TRUE
- Is Odd = FALSE
- Sign = -1



These numeric tools make data preparation and analysis easier by helping us perform calculations, handle numbers efficiently and get meaningful results in Power BI.

Date and Time tool

Date & Time tools in Power BI help us create and manipulate date and time values for better analysis and reporting.



1 Date

Date tool helps to extract different parts of a date, perform date calculations and find specific dates.

Function	What it does	Example (for 15-08-2024)
Age	Calculates age from a date of birth.	AGE(15-08-2000) → 24
Date only	Returns only the date part (from datetime).	DATE(2024,8,15)
Year	Extracts the year.	YEAR(15-08-2024) → 2024
Month	Extracts the month (1-12).	MONTH(15-08-2024) → 8
Quarter	Returns the quarter (1-4).	QUARTER(15-08-2024) → 3
Week	Returns the week number (of the year).	WEEKNUM(15-08-2024) → 33
Day	Extracts the day of month.	DAY(15-08-2024) → 15

Additional useful functions:

- Subtract days - DATEADD(date, -7, DAY)
- Combine date and time - COMBINE DATE(date, time)
- Earliest date - MIN(date column)
- Latest date - MAX(date column)

Example:

If we have a column [Order Date] and [Delivery Time], we can combine them:

Combined DateTime = COMBINE DATE([Order Date], [Delivery Time])

Order Date	Delivery Time	Combined DateTime
10-08-2024	14:30:00	10-08-2024 14:30:00
12-08-2024	09:15:00	12-08-2024 09:15:00

Remember:

- Date functions work with date values (not text).
- For text, convert using DATEVALUE() first.
- These tools help in grouping, filtering, and creating useful columns for visuals in Power BI.

2 Time

Time tool helps to extract and manipulate time values from a date-time column.

Function	What it does	Example (for 14:45:30)
Hour	Extracts the hour (0-23).	HOURL(14:45:30) → 14
Minute	Extracts the minute (0-59).	MINUTE(14:45:30) → 45
Second	Extracts the second (0-59).	SECOND(14:45:30) → 30
Time only	Returns only the time part.	TIME(14:45:30) → 14:45:30
AM/PM	Returns AM or PM.	FORMAT(14:45:30, "AM/PM") → PM
Add time	Adds time to a date/time.	DATEADD(datetime, 2, HOUR)

Example:

From a datetime column [LoginDateTime] we can get time parts:

LoginDateTime	Hour	Minute	Second	AM/PM
10-08-2024 09:23:45	9	23	45	AM
11-08-2024 17:05:12	17	5	12	PM

3 Duration

Duration tool helps to calculate the difference between two date/time values and returns the time span.

Function	What it does	Example
Duration (Days)	Difference in days.	DATEDIFF(StartDate, EndDate, DAY)
Duration (Hours)	Difference in hours.	DATEDIFF(StartDate, EndDate, HOUR)
Duration (Minutes)	Difference in minutes.	DATEDIFF(StartDate, EndDate, MINUTE)
Duration (Seconds)	Difference in seconds.	DATEDIFF(StartDate, EndDate, SECOND)
Total Duration	Returns total duration as a time value.	ENDTIME(StartDate, EndDate)

Example:

- Find the duration between Order and Delivery date:

Order DateTime	Delivery DateTime	Duration (Days)	Duration (Hours)
10-08-2024 10:00	12-08-2024 16:00	2	54
15-08-2024 09:30	16-08-2024 11:00	1	25.5

★ Use Case:

- Age for customer analysis
- Week/Quarter for trend analysis
- Duration for project/response time
- Combine date & time for accurate filtering

Locale & Rolling Calendar in Power BI



1. Locale

What is Locale?

- Locale is a regional setting in Power BI that tells the system how to interpret and display data like dates, numbers, currency, and text.
- It helps handle different regional formats (e.g. dd/mm/yyyy vs mm/dd/yyyy, different month names, decimal separators, etc.).
- It is especially important when working with date and time data, because the same date can be written in different formats in different countries.

Use of Locale with DATE

- The DATE function creates a date value based on year, month and day. The order of arguments and the display format depend on the locale.
- We can also use the locale to convert text dates into a proper date format using DATEVALUE().

Example 1: Different date formats using locale

Locale	DATE(2024, 8, 15)	Display Format
English (US) (en-US)	8/15/2024	MM/DD/YYYY
English (UK) (en-GB)	15/08/2024	DD/MM/YYYY
German (de-DE)	15.08.2024	DD.MM.YYYY
French (fr-FR)	15/08/2024	DD/MM/YYYY
Spanish (es-ES)	15/08/2024	DD/MM/YYYY

Example 2: Using locale with DATEVALUE()

Suppose we have a text date in "dd/mm/yyyy" format and we want to convert it to a date value.

Expression	Locale	Result
DATEVALUE("15/08/2024")	en-GB	15-Aug-2024
DATEVALUE("15/08/2024")	en-US	Aug 15, 2024 (interpreted wrongly)
DATEVALUE("08/15/2024")	en-US	15-Aug-2024

Key Points

- Use the correct locale to avoid date format issues.
- It ensures consistent data across different regions.
- Common locales: en-US, en-GB, de-DE, fr-FR, es-ES, etc.



Locale = Correct date format + accurate analysis

2. Rolling Calendar

What is Rolling Calendar?

- A rolling calendar is a dynamic date table that always shows the most recent N days/months/years based on the current date.
- It is useful for creating time intelligence calculations like last 7 days, last 30 days, year-to-date, etc.
- Instead of using a fixed date range, it automatically updates as time moves forward.

Example

Let's create a rolling calendar for the last 7 days. Assume today's date is 15-Aug-2024.

1. Create a date table using DAX:

```
RollingCalendar =  
VAR TodayDate = TODAY()  
RETURN  
ADDCOLUMNS (  
    CALENDAR ( TodayDate - 6, TodayDate ),  
    "IsRolling", 1  
)
```

This will create a date table from 9-Aug-2024 to 15-Aug-2024 (last 7 days).

2. Result (Rolling Calendar Table)

Date	IsRolling
09-08-2024	1
10-08-2024	1
11-08-2024	1
12-08-2024	1
13-08-2024	1
14-08-2024	1
15-08-2024	1

This table will always update to show the last 7 days relative to today's date.

How it helps?

- Used in time intelligence (e.g., Last 7 Days, Last 30 Days).
- Keeps reports dynamic without manually updating dates.
- Useful for dashboards that need recent data.

Use Case

- Daily sales report (last 7 days).
- Customer activity (last 30 days).
- Financial reporting (rolling 12 months).



Rolling Calendar = Dynamic dates + Always up to date + Powerful for time based analysis